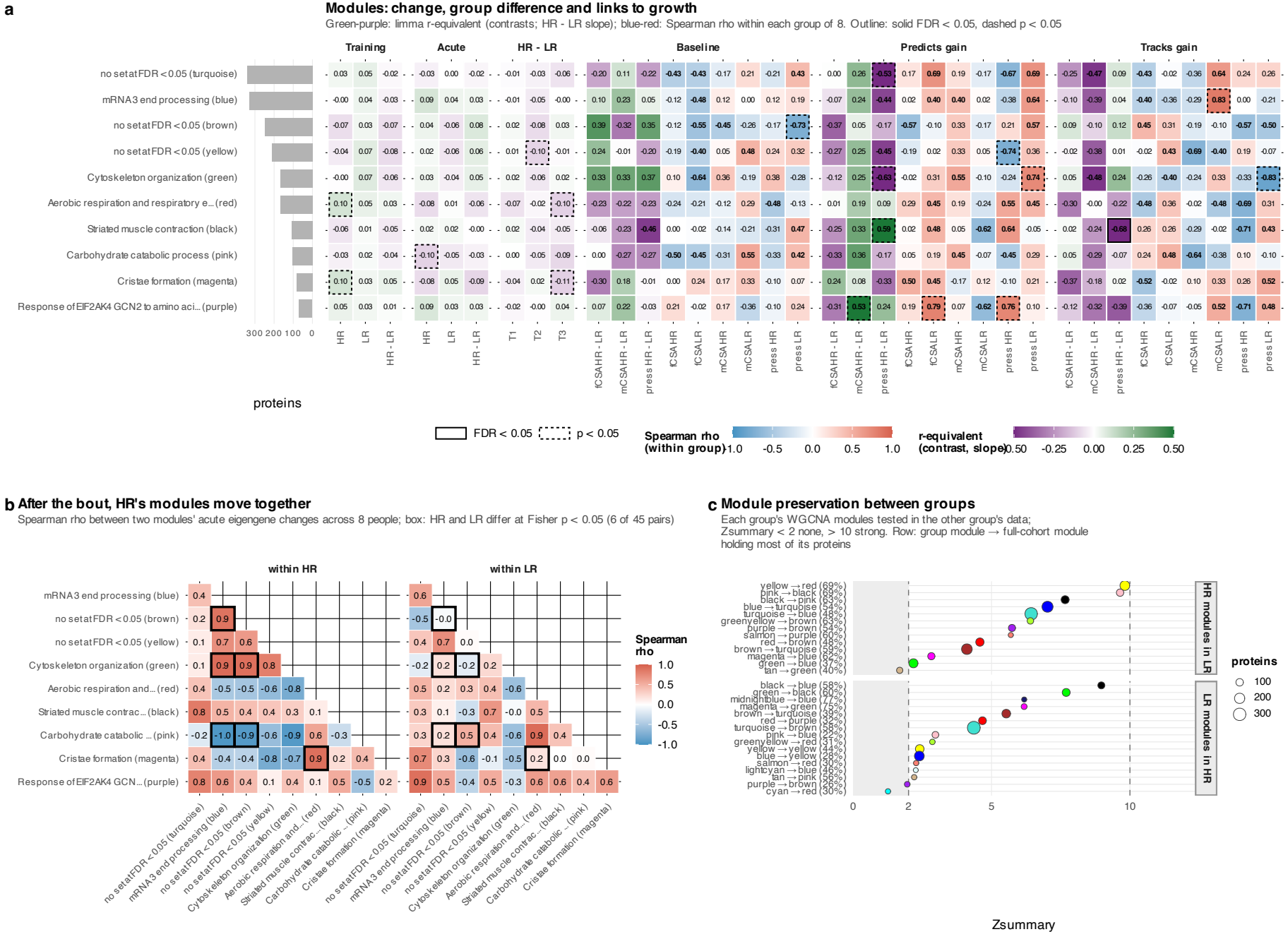
a
Modules: change, group difference and links to growth
Green-purple: limma r-equivalent (contrasts; HR - LR slope); blue-red: Spearman rho within each group of 8. Outline: solid FDR < 0.05, dashed p < 0.05
Training
Acute
HR - LR
Baseline
Predicts gain
Tracks gain
no set at FDR < 0.05 (turquoise)
mRNA3 end processing (blue)
no set at FDR < 0.05 (brown)
no set at FDR < 0.05 (yellow)
Cytoskeleton organization (green)
Aerobic respiration and respiratory e... (red)
Striated muscle contraction (black)
Carbohydrate catabolic process (pink)
Cristae formation (magenta)
Response of EIF2AK4 GCN2 to amino aci... (purple)
300 200 100 0
HR
LR
HR - LR
T1
T2
T3
fCSAHR - LR
mCSAHR - LR
press HR - LR
fCSAHR
fCSALR
mCSAHR
mCSALR
press HR
press LR
proteins
FDR < 0.05
p < 0.05
Spearman rho (within group)
1.0 -0.5 0.0 0.5 1.0
r-equivalent (contrast, slope)
0.50 -0.25 0.00 0.25 0.50
b After the bout, HR's modules move together
Spearman rho between two modules' acute eigengene changes across 8 people; box: HR and LR differ at Fisher p < 0.05 (6 of 45 pairs)
within HR
within LR
Spearman rho
1.0
0.5
0.0
-0.5
-1.0
c Module preservation between groups
Each group's WGCNA modules tested in the other group's data;
Zsummary < 2 none, > 10 strong. Row: group module → full-cohort module
holding most of its proteins
yellow → red (69%)
pink → black (69%)
black → pink (63%)
blue → turquoise (54%)
turquoise → blue (48%)
greenyellow → brown (63%)
purple → brown (54%)
salmon → purple (60%)
red → brown (48%)
brown → turquoise (59%)
magenta → blue (62%)
green → blue (37%)
tan → green (40%)
black → blue (58%)
green → black (60%)
midnightblue → blue (77%)
magenta → green (75%)
brown → turquoise (40%)
red → purple (32%)
turquoise → brown (58%)
pink → blue (22%)
greenyellow → red (31%)
yellow → yellow (44%)
blue → yellow (28%)
salmon → red (30%)
lightcyan → blue (46%)
tan → pink (56%)
purple → brown (26%)
cyan → red (30%)
HR modules in LR
LR modules in HR
proteins
100
200
300
0
2
5
10
Zsummary